A Reflection Witness for Unsaturation of Čirić Quasi-Contractions

Solution packet for arXiv:2103.15148, Open Problem 2

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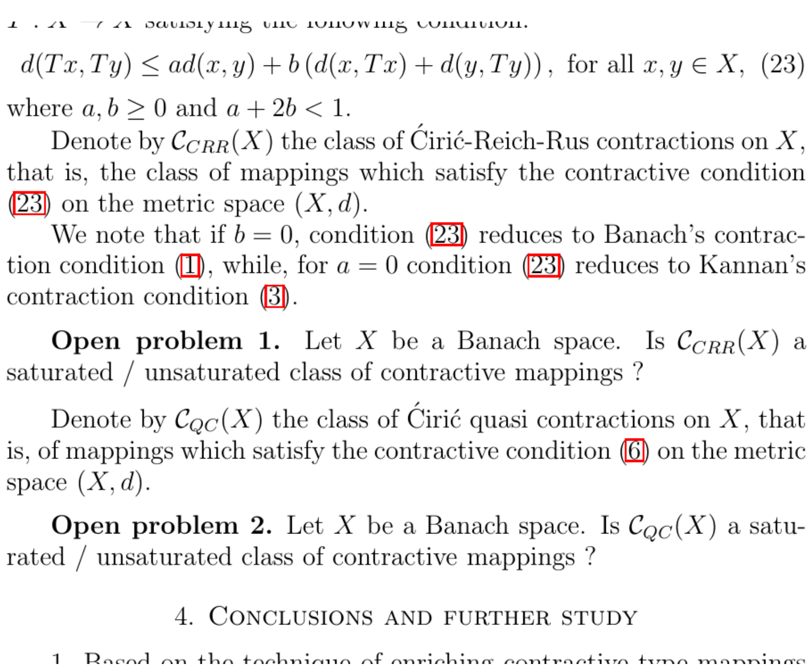
Status. Candidate full solution, likely valid. On every nonzero Banach space the class of Čirić quasi-contractions is unsaturated. The zero Banach space is the unique saturated edge case.

1 Source question

Berinde and Păcurar ask in Open Problem 2 on printed page 15 of [1] whether the class $\mathcal{C}_{\mathrm{QC}}(X)$ of Čirić quasi-contractions on a Banach space X is saturated or unsaturated. The source statement and its referenced definition are reproduced below.


$T : X \to X$ satisfying the following condition:

$$d(Tx, Ty) \le a d(x, y) + b\,(d(x, Tx) + d(y, Ty))\,, \text{ for all } x, y \in X, \quad (23)$$

where $a, b \ge 0$ and $a + 2b < 1$.

Denote by $\mathcal{C}_{CRR}(X)$ the class of Ćirić-Reich-Rus contractions on X, that is, the class of mappings which satisfy the contractive condition (23) on the metric space (X, d).

We note that if $b = 0$, condition (23) reduces to Banach's contraction condition (1), while, for $a = 0$ condition (23) reduces to Kannan's contraction condition (3).

Open problem 1. Let X be a Banach space. Is $\mathcal{C}_{CRR}(X)$ a saturated / unsaturated class of contractive mappings ?

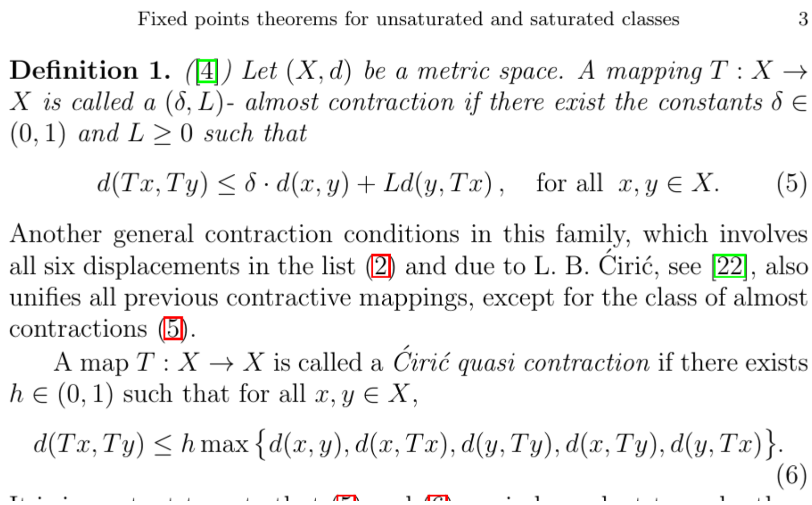
Denote by $\mathcal{C}_{QC}(X)$ the class of Ćirić quasi contractions on X, that is, of mappings which satisfy the contractive condition (6) on the metric space (X, d).

Open problem 2. Let X be a Banach space. Is $\mathcal{C}_{QC}(X)$ a saturated / unsaturated class of contractive mappings ?

4. Conclusions and further study

1. Based on the technique of enriching contractive type mappings

Figure 1: Open Problems 1 and 2 on printed page 15 of [1].


Fixed points theorems for unsaturated and saturated classes 3

Definition 1. *([4]) Let (X, d) be a metric space. A mapping $T : X \to X$ is called a (δ, L)- almost contraction if there exist the constants $\delta \in (0, 1)$ and $L \geq 0$ such that*

$$d(Tx, Ty) \leq \delta \cdot d(x, y) + Ld(y, Tx)\,, \quad \text{for all } \; x, y \in X. \tag{5}$$

Another general contraction conditions in this family, which involves all six displacements in the list (2) and due to L. B. Ćirić, see [22], also unifies all previous contractive mappings, except for the class of almost contractions (5).

A map $T : X \to X$ is called a *Ćirić quasi contraction* if there exists $h \in (0, 1)$ such that for all $x, y \in X$,

$$d(Tx, Ty) \leq h \max\big\{d(x, y), d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx)\big\}. \tag{6}$$

Figure 2: The defining inequality (6) on printed page 3 of [1].

For clarity, a self-map $T : X \to X$ is a *Čirić quasi-contraction* if there is $h \in (0, 1)$ such that

$$\|Tx - Ty\| \leq h \max\{\|x - y\|, \|x - Tx\|, \|y - Ty\|, \|x - Ty\|, \|y - Tx\|\} \tag{1}$$

for every $x, y \in X$. For $0 < \lambda \leq 1$, its averaged map is

$$T_\lambda = (1 - \lambda)I + \lambda T.$$

A map is $\mathcal{C}_{\mathrm{QC}}$-enriched if $T_\lambda \in \mathcal{C}_{\mathrm{QC}}(X)$ for some such λ. The class is *saturated* when $\mathcal{C}_{\mathrm{QC}}(X)^e = \mathcal{C}_{\mathrm{QC}}(X)$ and *unsaturated* when this inclusion is strict.

2 Idea of the proof

The averaging operation has a simple extreme behavior that the five-distance maximum in (1) cannot hide. Take the reflection $R = -I$. At the opposite pair $u, -u$, every nonzero term in the maximum equals $2\|u\|$, exactly the distance between the images. The strict factor $h < 1$ therefore makes the quasi-contraction inequality impossible. But averaging R with the identity at $\lambda = 1/2$ cancels it completely: $R_{1/2} = 0$, a constant map and hence a quasi-contraction. Thus one map lies in the enriched class but not in the original class.

3 Full answer

Theorem 1. *Let X be a real or complex Banach space.*

1. *If $X \neq \{0\}$, then $\mathcal{C}_{\mathrm{QC}}(X)$ is unsaturated.*
2. *If $X = \{0\}$, then $\mathcal{C}_{\mathrm{QC}}(X)$ is saturated.*

Consequently, every nonzero Banach space has the same answer to Open Problem 2 of [1]: the class is unsaturated.

Proof. Assume first that $X \neq \{0\}$, and let $R = -I_X$. Choose $u \neq 0$ and put $x = u$, $y = -u$. Then

$$\|Rx - Ry\| = 2\|u\|.$$

For this same pair, the five quantities in the maximum in (1) are, in order,

$$2\|u\|, \quad 2\|u\|, \quad 2\|u\|, \quad 0, \quad 0.$$

If R satisfied (1) for some $h < 1$, we would therefore have $2\|u\| \leq 2h\|u\|$, an impossibility. Hence $R \notin \mathcal{C}_{\mathrm{QC}}(X)$.

On the other hand,

$$R_{1/2} = \tfrac{1}{2}I_X + \tfrac{1}{2}(-I_X) = 0.$$

The zero map belongs to $\mathcal{C}_{\mathrm{QC}}(X)$: its left-hand side in (1) is zero for all x, y, so, for example, $h = 1/2$ works. Thus $R \in \mathcal{C}_{\mathrm{QC}}(X)^e \setminus \mathcal{C}_{\mathrm{QC}}(X)$, proving strict inclusion and unsaturation.

If $X = \{0\}$, there is only one self-map. It is the zero map, belongs to $\mathcal{C}_{\mathrm{QC}}(X)$, and every one of its averages is itself. Hence $\mathcal{C}_{\mathrm{QC}}(X)^e = \mathcal{C}_{\mathrm{QC}}(X)$. $\square$

The same witness also gives a one-line proof of the source's Open Problem 1, although that problem was independently answered in later literature [2].

Corollary 2. *For every nonzero Banach space X, the class $\mathcal{C}_{\mathrm{CRR}}(X)$ of Čirić–Reich–Rus contractions is unsaturated; it is saturated on the zero space.*

Proof. For $R = -I$ and $x = u$, $y = -u$, the defining right-hand side is

$$a\|x - y\| + b(\|x - Rx\| + \|y - Ry\|) = 2(a + 2b)\|u\| < 2\|u\| = \|Rx - Ry\|,$$

because $a, b \geq 0$ and $a + 2b < 1$. Thus $R \notin \mathcal{C}_{\mathrm{CRR}}(X)$, whereas $R_{1/2} = 0 \in \mathcal{C}_{\mathrm{CRR}}(X)$. The zero-space argument is unchanged. $\square$

4 Verification and limitations

The proof uses only the definitions. The critical checks are: $1/2$ is an allowed averaging parameter; the reflection is a self-map on every real or complex Banach space; the opposite pair exists exactly when the space is nonzero; and a constant map satisfies both defining inequalities. No completeness property beyond the source's Banach-space hypothesis is used, so the same result holds on every nonzero normed space.

The theorem classifies saturation only for the source's class $\mathcal{C}_{\mathrm{QC}}(X)$ and its exact enrichment operation. It does not claim new fixed point or iteration estimates for the larger enriched class.

5 Novelty check and review recommendation

A bounded search through August 11, 2026 used the exact phrases "Čirić quasi contractions saturated", "class of Čirić quasi contractions unsaturated", and the wording of Open Problem 2, together with searches for the source title and authors. It found the source and later general fixed-point papers, but no paper explicitly answering Open Problem 2. The same search found Berinde–Păcurar's 2022 paper [2], which explicitly answers Open Problem 1 for Čirić–Reich–Rus contractions. Accordingly, only Theorem 1 is presented as the candidate new full result.

Human review is recommended as a high-confidence, low-complexity check. The main point to verify is semantic: that Definition 2.1 of [1] allows the self-map $-I_X$ and $\lambda = 1/2$ exactly as used above.

References

[1] V. Berinde and M. Păcurar, *Fixed points theorems for unsaturated and saturated classes of contractive mappings in Banach spaces*, Symmetry **13** (2021), 713; arXiv:2103.15148.

[2] V. Berinde and M. Păcurar, *A new class of unsaturated mappings: Čirić–Reich–Rus contractions*, An. Univ. Ovidius Constanţa Ser. Mat. **30** (2022), no. 3, 37–50, doi:10.2478/auom-2022-0033.